

Lecture 2: Review of Probability Theory II

Lecturer: Ziqiao Wang

In this lecture¹, we will introduce several ways of specifying a discrete or continuous random variable or distribution, including the cumulative distribution function, probability mass function, and probability density function. We then cover functions of a random variable, the expected value and variance, and two important probability inequalities.

1 CDF, PMF and PDF

All events of interest on the real line can be expressed as sets of the form $\{\zeta : X(\zeta) \leq b\}$, $b \in \mathbb{R}$.

Definition 1.1 (Cumulative distribution function). The cumulative distribution function (CDF) of a random variable X is defined as the probability of the event $\{X \leq x\}$:

$$F_X(x) = P(X \leq x) \quad \text{for } -\infty < x < +\infty.$$

Clearly, $F_X(x)$ is a function of the variable x , since its value changes as x varies.

We are now ready to state the fundamental properties of the CDF. From the axioms of probability and their corollaries, it follows that the CDF satisfies the following properties:

- (i) $0 \leq F_X(x) \leq 1$.
- (ii) $\lim_{x \rightarrow \infty} F_X(x) = 1$.
- (iii) $\lim_{x \rightarrow -\infty} F_X(x) = 0$.
- (iv) $F_X(x)$ is a nondecreasing function of x , that is, if $a < b$, then $F_X(a) \leq F_X(b)$.
- (v) $F_X(x)$ is continuous from the right, that is, for $h > 0$,

$$F_X(b) = \lim_{h \rightarrow 0} F_X(b+h) = F_X(b^+).$$

- (vi) $\mathbb{P}[a < X \leq b] = F_X(b) - F_X(a)$.
- (vii) $\mathbb{P}[X = b] = F_X(b) - F_X(b^-)$.
- (viii) $\mathbb{P}[X > x] = 1 - F_X(x)$.

Remark 1.1. If the CDF is continuous at a point b , then $P(X = b) = 0$.

We may loosely say that a R.V. X is discrete if it takes values from a countable set, i.e. $\Omega_X = \{x_1, x_2, \dots\}$. A more formal definition is given below.

Definition 1.2 (Discrete R.V. and PMF). A discrete random variable X is defined as a random variable whose CDF is a right-continuous, staircase function of x , with jumps at a countable set of points $\{x_1, x_2, \dots\}$. The probability mass function (PMF) of a discrete random variable X is defined as:

$$p_X(x_k) \triangleq P(X = x_k) \quad \text{for } x_k \in \Omega_X.$$

¹Reading: Chapter 4-5 of Leon-Garcia.

Notably, the CDF of a discrete random variable is given by the cumulative probability of all outcomes less than x , and can be expressed as a weighted sum of unit step functions:

$$F_X(x) = \sum_k p_X(x_k)u(x - x_k),$$

where

$$u(x) = \begin{cases} 0, & \text{for } x < 0, \\ 1, & \text{for } x \geq 0, \end{cases}$$

is the unit step function.

Definition 1.3 (Continuous R.V. and PDF). If $F_X(x)$ is continuous everywhere on $\mathbb{R}$, then we say the random variable is a continuous random variable. The probability density function (PDF) of a continuous random variable X is defined as, for all $x \in \mathbb{R}$,

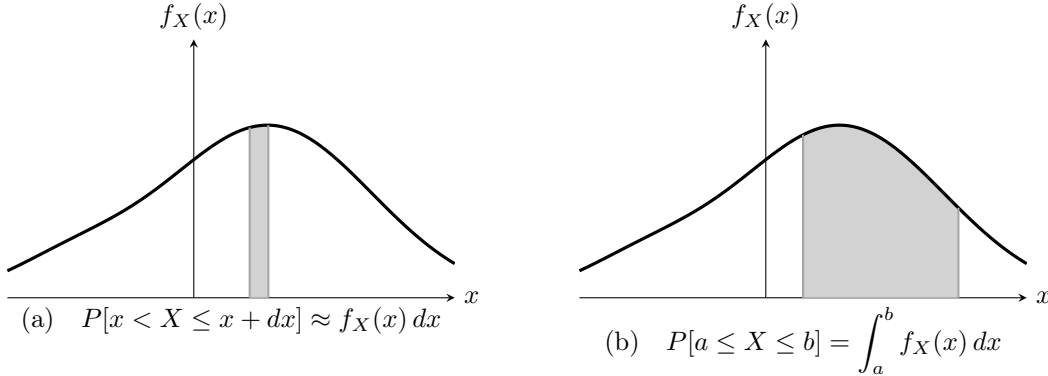
$$f_X(x) \triangleq \begin{cases} \frac{d}{dx}F_X(x), & \text{if } \frac{d}{dx}F_X(x) \text{ exists;} \\ 0, & \text{otherwise.} \end{cases}$$

Definition 1.4 (Mixed Typy R.V.). A random variable which is neither continuous nor discrete is termed a random variable of mixed type.

A mixed-type random variable is one whose CDF exhibits jumps at a countable set of points $\{x_1, x_2, \dots\}$ while also increasing continuously over at least one interval of x values. The CDF of such a random variable can be expressed in the following form:

$$F_X(x) = pF_1(x) + (1 - p)F_2(x),$$

where $p \in (0, 1)$, and $F_1(x)$ is the CDF of a discrete R.V. and $F_2(x)$ is the CDF of a continuous R.V.



PDF. The PDF represents the “density” of probability at a point x in the following sense: for a sufficiently small $h > 0$, the probability that X falls within a small interval around x , namely $[x, x + h]$, is approximately proportional to the value of the PDF at x .

$$P(x < X \leq x + h) = F_X(x + h) - F_X(x) = \frac{F_X(x + h) - F_X(x)}{h} h \simeq f_X(x)h.$$

The PDF satisfies the following properties:

- (i) $f_X(x) \geq 0$.
- (ii) $P(a \leq X \leq b) = \int_a^b f_X(x) dx$.

$$(iii) \quad F_X(x) = \int_{-\infty}^x f_X(t) dt.$$

$$(iv) \quad 1 = \int_{-\infty}^{+\infty} f_X(t) dt.$$

The PDF completely specifies the behavior of continuous random variables.

2 Functions of a Random Variable

Let X be a R.V. and $g(x)$ a real-valued function defined on $\mathbb{R}$. Define $Y = g(X)$, then Y is itself a R.V., and the probabilities associated with the possible values of Y are determined by the mapping $g(x)$.

The CDF of Y is defined as the probability of the event $\{Y \leq y\}$. In principle, it can be obtained by evaluating the probability of the equivalent event $\{g(X) \leq y\}$.

Example 1 (A Linear Function). Let the random variable Y be defined by

$$Y = aX + b,$$

where a is a nonzero constant. Suppose that X has CDF $F_X(x)$. Find $F_Y(y)$.

The event $\{Y \leq y\}$ occurs when $A = \{aX + b \leq y\}$ occurs. If $a > 0$, then $A = \{X \leq (y - b)/a\}$, and thus

$$F_Y(y) = P\left(X \leq \frac{y - b}{a}\right) = F_X\left(\frac{y - b}{a}\right), \quad a > 0.$$

On the other hand, if $a < 0$, then $A = \{X \geq (y - b)/a\}$, and

$$F_Y(y) = P\left(X \geq \frac{y - b}{a}\right) = 1 - F_X\left(\frac{y - b}{a}\right), \quad a < 0.$$

We can obtain the PDF of Y by differentiating with respect to y . Using the chain rule,

$$\frac{dF}{dy} = \frac{dF}{du} \frac{du}{dy},$$

where u is the argument of F . In this case, $u = (y - b)/a$. Hence, for $a > 0$,

$$f_Y(y) = \frac{1}{a} f_X\left(\frac{y - b}{a}\right), \quad a > 0,$$

and for $a < 0$,

$$f_Y(y) = \frac{1}{-a} f_X\left(\frac{y - b}{a}\right), \quad a < 0.$$

The above two results can be written compactly as

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y - b}{a}\right). \quad (1)$$

Example 2 (A Linear Function of a Gaussian Random Variable). Let X be a R.V. with a Gaussian PDF with mean m and standard deviation σ :

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-(x-m)^2/2\sigma^2}, \quad -\infty < x < \infty. \quad (2)$$

Let $Y = aX + b$, then find the PDF of Y . Substitution of Eq. (2) into Eq. (1) yields

$$f_Y(y) = \frac{1}{\sqrt{2\pi} |a\sigma|} e^{-(y-b-am)^2/2(a\sigma)^2}.$$

Note that Y also has a Gaussian distribution with mean $b + am$ and standard deviation $|a|\sigma$. Therefore, *a linear function of a Gaussian R.V. is also a Gaussian R.V..*

3 Expectation and Variance

Definition 3.1 (Expectation). The expected value or mean of a random variable X is defined by

$$\mathbb{E}[X] \triangleq \int_{-\infty}^{+\infty} t f_X(t) dt$$

Expectation is a linear operator:

- (i) $\mathbb{E}[g(X) + h(X)] = \mathbb{E}[g(X)] + \mathbb{E}[h(X)]$.
- (ii) $\mathbb{E}[aX] = a\mathbb{E}[X]$.
- (iii) $\mathbb{E}[X + c] = \mathbb{E}[X] + c$.

Proposition 1 (Mean of a symmetric distribution). *Let X be a continuous random variable with density f_X and $\mathbb{E}[|X|] < \infty$. Suppose that f_X is symmetric about $m \in \mathbb{R}$, namely,*

$$f_X(m - x) = f_X(m + x), \quad \text{for all } x \in \mathbb{R}.$$

Then $\mathbb{E}[X] = m$.

Proof. First, observe that

$$\int_{-\infty}^{\infty} |m - t| f_X(t) dt \leq |m| + \mathbb{E}[|X|] < \infty,$$

so the following integrals are absolutely convergent.

Splitting the integral at $t = m$ and using the substitutions $x = m - t$ on the left and $x = t - m$ on the right, we obtain

$$\begin{aligned} \int_{-\infty}^{\infty} (m - t) f_X(t) dt &= \int_{-\infty}^m (m - t) f_X(t) dt + \int_m^{\infty} (m - t) f_X(t) dt \\ &= \int_0^{\infty} x f_X(m - x) dx - \int_0^{\infty} x f_X(m + x) dx \\ &= \int_0^{\infty} x [f_X(m - x) - f_X(m + x)] dx \\ &= 0, \end{aligned}$$

where the last equality follows from symmetry.

On the other hand,

$$\int_{-\infty}^{\infty} (m - t) f_X(t) dt = m \int_{-\infty}^{\infty} f_X(t) dt - \int_{-\infty}^{\infty} t f_X(t) dt = m - \mathbb{E}[X].$$

Therefore, $\mathbb{E}[X] = m$. □

Expectation via tail probabilities. Let X be a nonnegative continuous random variable with CDF F_X . Then

$$\mathbb{E}[X] = \int_0^{\infty} (1 - F_X(t)) dt = \int_0^{\infty} \mathbb{P}(X > t) dt. \quad (3)$$

Proof. For every $x \geq 0$, we have

$$x = \int_0^{\infty} \mathbf{1}_{\{x > t\}} dt,$$

since the indicator equals one precisely when $0 \leq t < x$.

Applying this identity to X and taking expectations yields

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}\left[\int_0^\infty \mathbf{1}_{\{X>t\}} dt\right] = \int_0^\infty \mathbb{E}[\mathbf{1}_{\{X>t\}}] dt = \int_0^\infty \mathbb{P}(X > t) dt \\ &= \int_0^\infty (1 - F_X(t)) dt.\end{aligned}$$

The interchange of expectation and integration is justified by Tonelli's theorem, since the integrand is nonnegative. The identity remains valid when both sides equal $+\infty$. $\square$

Definition 3.2 (Variance). The variance of the random variable X is defined by

$$\text{Var}[X] \triangleq \mathbb{E}_X \left[(X - \mathbb{E}[X])^2 \right] = \mathbb{E}[X^2] - \mathbb{E}^2[X].$$

The standard deviation of the R.V. X is defined by

$$\text{Std}[X] \triangleq (\text{Var}[X])^{1/2}.$$

$$(i) \text{ Var}[X + c] = \text{Var}[X].$$

$$(ii) \text{ Var}[cX] = c^2 \text{Var}[X].$$

Definition 3.3 (Moments). The n th moment of the R.V. X is defined by

$$\mathbb{E}[X^n] \triangleq \int_{-\infty}^{+\infty} x^n f_X(x) dx.$$

4 Two Probability Inequalities

Theorem 4.1 (Markov's Inequality). If X is a nonnegative R.V. with mean $\mathbb{E}[X]$, then for any $a > 0$

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof of Markov's inequality. Let X be a nonnegative R.V. with $\mathbb{E}[X] < \infty$, and let $a > 0$. Since

$$X \geq a \mathbf{1}_{\{X \geq a\}} \quad \text{almost surely,}$$

taking expectations on both sides gives

$$\mathbb{E}[X] \geq a \mathbb{E}[\mathbf{1}_{\{X \geq a\}}] = a \mathbb{P}(X \geq a).$$

Dividing by $a > 0$, we obtain

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

This completes the proof. $\square$

Alternative proof of Markov's inequality via tail probabilities. Let X be a nonnegative R.V. with $\mathbb{E}[X] < \infty$, and let $a > 0$. By the tail-integral formula,

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t) dt.$$

For every $0 \leq t < a$, we have

$$\{X \geq a\} \subseteq \{X > t\},$$

and hence

$$\mathbb{P}(X > t) \geq \mathbb{P}(X \geq a).$$

Consequently,

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t) dt \geq \int_0^a \mathbb{P}(X > t) dt \geq \int_0^a \mathbb{P}(X \geq a) dt = a \mathbb{P}(X \geq a).$$

The value of the integrand at the single endpoint $t = a$ does not affect the integral. Dividing by $a > 0$ yields

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Geometrically, $\mathbb{E}[X]$ is the total area under the tail-probability curve $t \mapsto \mathbb{P}(X > t)$. Over $0 \leq t < a$, this curve lies above a rectangle of width a and height $\mathbb{P}(X \geq a)$. Thus, the rectangle's area, $a \mathbb{P}(X \geq a)$, cannot exceed $\mathbb{E}[X]$. $\square$

Theorem 4.2 (Chebyshev's Inequality). *Let X be a R.V. with variance σ^2 and expected value m , then for any $a \geq 0$, we have*

$$\mathbb{P}(|X - m| \geq a) \leq \frac{\sigma^2}{a^2}.$$

Proof of Chebyshev's inequality. Let X be a random variable with mean $m = \mathbb{E}[X]$ and finite variance $\sigma^2 = \mathbb{E}[(X - m)^2]$. For any $a > 0$, observe that

$$\{|X - m| \geq a\} = \{(X - m)^2 \geq a^2\}.$$

Applying Markov's inequality to the nonnegative random variable $(X - m)^2$ with threshold a^2 , we obtain

$$\begin{aligned} \mathbb{P}(|X - m| \geq a) &= \mathbb{P}((X - m)^2 \geq a^2) \\ &\leq \frac{\mathbb{E}[(X - m)^2]}{a^2} \\ &= \frac{\sigma^2}{a^2}. \end{aligned}$$

Therefore,

$$\mathbb{P}(|X - m| \geq a) \leq \frac{\sigma^2}{a^2}.$$

This completes the proof. $\square$

Remark 4.1 (Intuition behind the two inequalities). *Both inequalities express the same principle: large values cannot occur too frequently when an appropriate average is small.*

Markov's inequality: the mean controls large values. *Suppose that $X \geq 0$ and the event $\{X \geq a\}$ has probability p . On this event, X is at least a , so these outcomes alone contribute at least ap to $\mathbb{E}[X]$. The remaining outcomes contribute nonnegative amounts. Consequently, $ap \leq \mathbb{E}[X]$: a nonnegative random variable cannot have both a small mean and a high probability of taking very large values. For example, if a nonnegative waiting time has mean 10 minutes, then*

$$\mathbb{P}(X \geq 50) \leq \frac{10}{50} = 0.2.$$

Chebyshev's inequality: the variance controls deviations from the mean. *Variance is the average squared distance from the mean. Every outcome satisfying $|X - m| \geq a$ contributes a squared deviation of at least a^2 . Thus, a small variance leaves little room for large deviations to occur frequently. This is precisely Markov's inequality applied to the nonnegative random variable $(X - m)^2$.*

When $\sigma > 0$, setting $a = k\sigma$ gives

$$\mathbb{P}(|X - m| \geq k\sigma) \leq \frac{1}{k^2}, \quad k > 0.$$

For example,

$$\mathbb{P}(|X - m| < 2\sigma) \geq \frac{3}{4}, \quad \mathbb{P}(|X - m| < 3\sigma) \geq \frac{8}{9}.$$

These guarantees hold for every distribution with finite variance, not just Gaussian distributions. If $\sigma = 0$, then $X = m$ almost surely.

An upper bound is not an exact probability. These inequalities require no knowledge of the full distribution beyond their stated assumptions, which makes them broadly applicable but potentially conservative. The actual tail probability may be much smaller than the bound. Moreover, a bound greater than or equal to 1 gives no additional information beyond the fact that probabilities cannot exceed 1.

In short, Markov's inequality controls the magnitude of a nonnegative random variable, whereas Chebyshev's inequality controls its fluctuations around the mean.